Mohammad **Sarabian**

Modeling & Simulation Scientist at W. L. Gore & Associates

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Summary

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Innovative computational scientist with **Ph.D. in Mechanical Engineering** and over 5 years of combined postdoctoral and industry experience developing cutting-edge **AI/ML algorithms for medical imaging applications**. Currently serving as Modeling & Simulation Scientist at W. L. Gore & Associates, where I lead the development of novel computational frameworks and AI solutions for complex biomedical challenges, directly contributing to improved patient outcomes and next-generation medical device innovation.

Proven expertise in **medical image analysis** and deep learning architectures, including Physics-Informed Neural Networks (PINNs), transformers, and hybrid modeling approaches for clinical applications. Expert in applying state-of-the-art ML techniques (CNNs, transformers, VAEs, GANs, reinforcement learning) to multimodal medical datasets including **MRI, angiography, and Doppler ultrasound**. Strong publication record with first-author papers in high-impact journals (*IEEE Transactions on Medical Imaging*, *Journal of Fluid Mechanics*) and active presentation at major scientific conferences (APS-DFD). Proficient in **Python, MATLAB, Java**, and high-performance computing systems with deep expertise in developing **scalable algorithms from research concept to clinical implementation**.

Strong background in interdisciplinary collaboration across engineering and biomedical domains, with demonstrated ability to drive independent research projects while fostering cross-functional partnerships. Experienced in developing **automated diagnostic frameworks, deformable image registration (DIR)** algorithms, especially **DiffuseMorph**, and AI solutions that streamline clinical workflows. Track record of innovation evidenced by **patent applications** and open-source projects spanning medical imaging, numerical methods, and scientific ML applications.

Additional differentiating expertise: Unique combination of rigorous physics-based modeling foundation with advanced AI/ML capabilities positions me to develop novel algorithms that bridge academic research with clinical practice, essential for advancing medical imaging solutions in vendor-neutral platforms serving diverse healthcare organizations.

Experience

W. L. Gore & Associates — *Phoenix, Arizona, USA*

Modeling and Simulation Scientist — *Mar. 2023 - Present*

- Spearhead the development and implementation of cutting-edge AI-driven generative computational fluid dynamics (CFD) models and novel numerical techniques to simulate complex hemodynamics, optimize medical devices, and advance personalized medicine approaches, directly contributing to improved patient outcomes and next-generation healthcare solutions

OriGen.AI, Inc. — *New York, New York, USA*

Artificial Intelligence (AI) Researcher — *Mar. 2022 - Mar. 2023*

- Pioneered the development of a novel physics-inspired artificial intelligence (AI) framework to revolutionize computational fluid dynamics (CFD) simulations of multiphase porous media, significantly accelerating processing times while maintaining high accuracy in complex geological models
- Engineered cutting-edge physics-informed surrogate models to dramatically enhance the efficiency and precision of assisted history matching (AHM) in reservoir characterization, integrating advanced machine learning techniques to optimize oil and gas exploration and production strategies

University of Arizona — *Tucson, Arizona, USA*

Postdoctoral Research Associate in Biomedical Engineering — *Mar 2020 - Mar. 2022*

- Developed a pioneering deep learning model for automated diagnosis and quantification of cerebrovascular diseases, enhancing diagnostic accuracy and speed
- Engineered an innovative mathematical and CFD model for automatic generation of brain perfusion maps from MRI data
- Created a novel physics-inspired AI model to characterize human brain properties
- Mentored graduate students in cutting-edge biomedical engineering projects, fostering next-generation research talent

Education

PhD. in Mechanical Engineering
2015- Dec 2020

Ohio University, Athens, Ohio , USA

- Dissertation: *"Experimental Investigations and Direct Numerical Simulations of Rigid Particles in Shear Flows of Newtonian and Complex Fluids"*
- Advisor: **Dr. Sarah Hormozi**; Ohio University, Athens, Ohio, USA
- Co-Advisor: **Dr. Luca Brandt**; KTH Royal Institute of Technology, Stockholm, Sweden
- Co-Advisor: **Dr. Bloen Metzger**; Aix Marseille Université-IUSTI CNRS, Marseille, France
- GPA: 4.0

MS. in Mechanical Engineering
2012- Sep 2014

Shiraz University, Shiraz, Iran

- Thesis: *"Investigating methods for reduction of detailed combustion mechanisms of light hydrocarbons and hydrogen."*
- Advisor: **Dr. MohammadHadi Akbari**
- GPA: 4.0

BS. in Mechanical Engineering
2007- Jul 2011

Shiraz University, Shiraz, Iran

- Senior Project: *"Numerical study of tip clearance & solidity on performance of a transonic compressor rotor using ANSYS/CFX."*
- Advisor: **Dr. Hossein Khaleghi**
- GPA: 3.5

Publications

- **Sarabian, M.**, De Niz, L., Sastry, S., & Abramowitz, S. (2025). High-fidelity numerical simulations of complex venous stenting: Comparative assessment of venous stent configurations in the Inferior Vena Cava. Manuscript submitted to *Journal of Vascular Surgery: Venous and Lymphatic Disorders*.
- Ashtiani, S. Z. **Sarabian, M.**, Laksari, K., & Babaee, H. (2024). Reconstructing blood flow in data-poor regimes: a vasculature network kernel for Gaussian process regression. *Journal of The Royal Society Interface* 21(217) 20240194. doi: 10.1098/rsif.2024.0194 https://doi.org/10.1098/rsif.2024.0194.
- Kamali, A*., **Sarabian, M*.**, & Laksari, K. (2023). Elasticity Imaging Using Physics-informed Neural Networks: Spatial Discovery of Elastic Modulus and Poisson's Ratio. *Acta Biomaterialia* 1742-7061. https://doi.org/10.1016/j.actbio.2022.11.024.
- **Sarabian, M.**, Rosti, M.E., & Brandt, L. (2023). Interface-resolved simulations of confinement effect on sedimentation of a sphere settling in yield-stress fluids. *Journal of non-Newtonian Fluid Mechanics* 104787. https://doi.org/10.1016/j.jnnfm.2022.104787.
- Rodriguez-Torrado, R., Pumar-Jimenez A., Ruiz-Mataran P., **Sarabian, M.**, Togelius J., Agudelo L., Rueda A. *et al* (2022). Geological Neural Network Methodology for Automatic History Match; Real Case for Rubiales Field. *In SPE Annual Technical Conference and Exhibition. OnePetro* https://doi.org/10.2118/210133-MS.
- **Sarabian, M.**, Babaee, B., & Laksari, K. (2022). Physics-informed neural networks for brain hemodynamic predictions using medical imaging. *IEEE Transactions on Medical Imaging* 41(9) 2285-2303. doi: 10.1109/TMI.2022.3161653 https://ieeexplore.ieee.org/abstract/document/9740143.
- **Sarabian, M.**, Babaee, B., & Laksari, K. (2021). Physics-informed neural networks for improving cerebral hemodynamics predictions. *arXiv preprint arXiv:2108.11498* https://arxiv.org/abs/2108.11498.
- Rashedi, A., **Sarabian, M.**, Firouznia, M., Roberts, D., Ovarlez, G., Hormozi, S. (2020). Shear-induced migration and axial development of particles in channel flows of non-Brownian suspensions. *AIChE J.* 66(12), e17100. https://doi.org/10.1002/aic.17100.
- **Sarabian, M.**, Rosti, M.E., Brandt, L., & Hormozi, S. (2020). Numerical simulations of a sphere settling in simple shear flows of yield stress fluids. *Journal of Fluid Mechanics* 896, A17 https://doi.org/10.1017/jfm.2020.316.
- **Sarabian, M.**, Firouznia, M., Metzger, B., & Hormozi, S. (2019). Fully developed and transient concentration profiles of particulate suspensions sheared in a cylindrical Couette cell. *Journal of Fluid Mechanics* 862, 659-671 https://doi.org/10.1017/jfm.2018.982.
- [**Cover**] Izbassarov, D., Rosti, M. E., Ardekani, M. N., **Sarabian, M.**, Hormozi, S., Brandt, L., & Tammisola, O. (2018). Computational modeling of multiphase viscoelastic and elastoviscoplastic flows. *International Journal for Numerical Methods in Fluids*, 88(12), 521-543 https://doi.org/10.1002/fld.4678.
- **Sarabian, M.** & Sastry, S. (2025). *Deep Reinforcement Learning and Spatiotemporal Transformers for Intelligent Drug Delivery in Thrombosis Prevention*. Manuscript in preparation.
- **Sarabian, M.**, Metzger, B., & Hormozi, S. (2022). *Wall-effect on migration of particulate suspensions*. Manuscript in preparation.

Patents

- Rodriguez-Torrado, R., **Sarabian, M.**. 2022. PhyDarcyNet: Physics-informed deep convolutional neural networks for learning 3D transient Darcy flows in heterogeneous porous media. *US Provisional Patent* 63/381, 872, filed on November 1, 2022.
- **Sarabian, M.**, Laksari, K. 2021. Diagnosis and detection of cerebral vasospasm using deep machine learning. *US Provisional Patent* filed on March 1, 2021.

Technical Skills

Medical Imaging and AI/ML Algorithms

- **Medical Image Analysis:** Advanced processing of **MRI, angiography, CT**, and Doppler ultrasound data for 3D anatomical modeling and clinical decision support
- **Registration and Segmentation:** Expert in developing **deformable image registration algorithms, *e.g., DiffuseMorph***, automated segmentation frameworks, and physics-informed approaches for radiation oncology applications
- **Deep Learning for Medical Data:** Convolutional Neural Networks (CNN), Transformers, Variational Autoencoders (VAE), Generative Adversarial Networks (GAN), Diffusion Models for clinical workflow optimization
- **Physics-Informed Neural Networks (PINNs):** Advanced development and implementation for medical device modeling, tissue characterization, and hemodynamic predictions
- **Reinforcement Learning:** Deep Q-Networks, Policy Gradient Methods, SAC for intelligent control systems and clinical decision support
- **ML Frameworks: PyTorch (advanced)**, TensorFlow/Keras

Numerical Methods and Scientific Computing

- **Advanced Numerical Methods:** Finite Volume Method, Finite Element Method, Immersed Boundary Method, numerical optimization for algorithm development
- **Mathematical Modeling:** Strong foundation in applied mathematics, statistics, and data management for solving complex research problems
- **Computational Fluid Dynamics (CFD):** Expert in hemodynamics simulation, complex flow modeling, and device performance optimization
- **Multiphysics Simulation:** Mechanical, thermal, and fluid flow modeling via coupling STAR-CCM+ with Abaqus

Programming and Software Development

- **Programming Languages: Python (advanced)**, **MATLAB**, **Java**, Fortran for scientific and engineering applications
- **Data Analysis & Visualization:** Advanced preprocessing, feature engineering, statistical analysis for clinical datasets
- **Cloud Computing Platforms:** AWS, Microsoft Azure, Google Cloud, Azure Machine Learning (AML)
- **High Performance Computing (HPC):** Scalable systems, distributed computing, parallel processing for large-scale simulations
- **Version Control & Automation:** Git, GitHub, software development best practices

Medical Device Development and Clinical Implementation

- **Engineering Software:** Simcenter STAR-CCM+, COMSOL, ANSYS (Workbench, Fluent, CFX, SpaceClaim), SimVascular, Abaqus
- **Algorithm Translation:** Proven ability to guide algorithms from conception to clinical practice, with track record of clinical implementation
- **Patient-Specific Modeling:** Processing medical imaging into 3D models for personalized treatment scenario planning
- **Clinical Dataset Analysis:** Extensive experience leveraging biological and medical datasets for predictive modeling
- **Statistical Shape Modeling:** Techniques for anatomical modeling and patient-specific applications
- **Patent Development:** Experience in generation and development of patent applications for medical imaging technologies

Research and Leadership Skills

- **Independent Research:** Demonstrated ability to manage research projects with substantial independence, define resource needs, execute experiments, and interpret results
- **Technical Documentation:** Creation and maintenance of technical documentation for research methods and results
- **Cross-Functional Collaboration:** Leading interdisciplinary teams across engineering, clinical science, and product development domains
- **Scientific Communication:** Publications in peer-reviewed journals, conference presentations, clear communication with stakeholders
- **Mentorship:** Experience serving as technical resource to colleagues and mentor to junior staff
- **Academic Engagement:** Active follow of academic literature in medical imaging, computer vision, and AI/ML research areas

Workshops and Training (Selected)

- **Reinforcement learning** beginner to master-AI in Python by Udemy. Certificate earned at August 9, 2022
- Simulation using **ANSYS-Fluent** English Version by Udemy. Certificate earned at May 15, 2021
- **Deep Learning** by deeplearning.ai on Coursera. Certificate earned at March 19, 2020
- **Machine Learning** by Stanford University on Coursera. Certificate earned at Friday, January 31, 2020 9:35 PM GMT
- Fluent, Mimix and ICEM; Mechanical Engineering scientific association of Shiraz University, Shiraz, Iran, May , 2014
- Solidworks; Mechanical Engineering scientific association of Shiraz University, Shiraz, Iran , May , 2012

Research Experiences (Selected)

Modeling & Simulation Scientist
Mar 2023-present

W. L. Gore & Associates, Arizona, USA

- Developed novel **intelligent drug delivery control** system combining **Deep Reinforcement Learning (SAC)** and **spatiotemporal transformers** for optimal anticoagulant control in microfluidic **thrombosis prevention**, achieving $> 1000\times$ speedup over traditional CFD approaches while maintaining high accuracy in predicting hemodynamic evolution and enabling real-time clinical applications `https://schedule.aps.org/dfd/2025/events/U25/9`
- Built an intelligent research assistant with orchestrated **AI agents** (planner, search, writer, email) that automates end-to-end research workflows, reducing manual research time by 80% | [GitHub] (`https://github.com/sohrabsarabian/deep-research-agentic-ai`) | [Demo] (`https://huggingface.co/spaces/SkepticML/smart_deep_research`)
- Developed a **hybrid Python–Java framework** for calibrating Auricchio-style **superelastic NiTi with JPype coupling**; achieved 16× speedups over pure-Python and packaged the workflow in a reproducible **Docker + Streamlit stack for one-command deployment.**
- Designed & developed advanced **AI-driven** CFD surrogate model, called *"FV-FluidAttentionNet"* to optimize medical device design, enhancing functionality and patient outcomes `https://meetings.aps.org/Meeting/DFD24/Session/ZC12.7`
- Formulated innovative mathematical models for rapid evaluation of medical device performance, accelerating development cycles
- Developed DreamForge: A cutting-edge AI-powered tool that transforms text prompts into video sequences by leveraging the power of CLIP and Taming Transformer models `https://github.com/sohrabsarabian/DreamForge-Text-to-Video-AI-Generation`
- Developed MiniLM: A Compact Language Model that offers a streamlined implementation of a small-scale Language Model (LM) `https://github.com/sohrabsarabian/MiniLM-A-Compact-Language-Model-Playground`
- Developed FaceForge: Advanced GAN techniques including self-attention mechanisms and residual blocks for Photorealistic Face Synthesis `https://github.com/sohrabsarabian/FaceForge-Advanced-GAN-for-Photorealistic-Face-Synthesis`
- Develped SARSA-DeepAscent: a powerful reinforcement learning technique that combines the SARSA (State-Action-Reward-State-Action) algorithm with deep neural networks `https://github.com/sohrabsarabian/SARSA-DeepAscent`
- Developed parallel framework for simulating 3D lid-driven cavity flow in Java `https://github.com/sohrabsarabian/CFDParallelJava3D`

Artificial Intelligence (AI) Researcher
Mar 2022-Mar 2023

Origen.AI, Inc, New York, USA

- Developed novel **physics-inspired artificial intelligence (AI)** framework to accelerate CFD simulations of multiphase porous media
- Developed unique **physics-informed surrogate models** for rapidly perform assisted history matching (AHM)
- Developed CNN+Transformer networks for improving the accuracy of **surrogate models**

Postdoctoral Research Associate
Mar 2020-present

Department of Biomedical Engineering, University of Arizona, AZ, USA

- Automatic diagnosis and quantification of cerebrpvascular diseases via developing novel **machine learning algorithms**
- Multi-modality brain haemodynamic analysis: *in vivo* 4D flow MRI, *in silico* **computational fluid dynamics** and **deep learning**
- Develop **deep physics-informed neural networks** for estimating human brain tissue mechanical properties `https://github.com/sohrabsarabian/PINN-Elasticity-Imaging`

Research Assistant
2015- Mar 2020

Department of Mechanical Engineering, Ohio University, OH, USA

- Developed customized Particle Image Velocimtery (PIV) & Particle Tracking Velocimetry (PTV) systems for dense suspensions from scratch in a curved Couette device
- Analyzed flow field around particles in shear flows of Newtonian and yield-stress by the means of Particle Image Velocimetry (PIV)
- Investigated motion of rigid particles in shear flows of Newtonian fluids using Particle Tracking Velocimetry (PTV)
- Simulated flow around a sphere in a viscoelastic fluid using FENE-P constitutive equation employing Immersed Boundary Method (IBM); collaboration with professor Luca Brandt, KTH Royal Institute of Technology
- Simulated the interaction of two-equally solid sphere in a confined shear flow of a viscoelastic fluid using Oldroyd-B model employing Immersed Boundary method (IBM) collaboration with professor Luca Brandt, KTH Royal Institute of Technology
- Performed precise thermal-assisted refractive index matching methods for high quality visualization of dense suspensions
- Characterized shear-induced migration of noncolloidal particles in Newtonian fluid via Refractive Index Matching Method (RIM)
- Simulated the viscoelasticity-induced migration of rigid solid spherical particles in a confined shear flow using Immersed Boundary Method (IBM) collaboration with professor Luca Brandt, KTH Royal Institute of Technology

Research Assistant
2007-2014

Department of Mechanical Engineering, Shiraz University, Shiraz, Iran

- Simulated a transonic compressor rotor using ANSYS/CFX commercial package
- Investigated the performance of a transonic compressor rotor by altering the solidity and tip clearance using ANSYS/TurboGrid & ANSYS/CFX commercial package
- Simulated auto-ignition process via coding with FORTRAN
- Simulated a microchannel water-gas shift surface reactor in 3D via coding with FORTRAN

Talks and Presentations (Selected)

- **Sarabian, M.**, & Sastry, S. (2025, November). *Deep Reinforcement Learning and Spatiotemporal Transformers for Intelligent Drug Delivery & Flow Control in Thrombosis Prevention*. The 78th Annual Meeting of the American Physical Society's Division of Fluid Dynamics (APS-DFD), Houston, TX.
- **Sarabian, M.**, & Sastry, S. (2024, November). *FV-FluidAttentionNet: A Label-Free Physics-Informed Autoencoder with Finite-Volume Discretization for Rapid Navier-Stokes Solutions*. The 77th Annual Meeting of the American Physical Society's Division of Fluid Dynamics (APS-DFD), Salk Lake City, Utah.
- **Sarabian, M.**, Rosti, M.E., & Brandt, L. (2023, November). *Particle-resolved simulations of the wall effect on the sedimentation of a single sphere in yield-stress fluids*. The 76th Annual Meeting of the American Physical Society's Division of Fluid Dynamics (APS-DFD), Washington, DC.
- **Sarabian, M.**, Mataran, P., & Torrado, R. (2022, November). *Finite PINN Net: Physics-informed deep convolutional neural networks for learning 3D transient Darcy flows in heterogeneous porous media*. The 75th Annual Meeting of the American Physical Society's Division of Fluid Dynamics (APS-DFD), Indianapolis, Indiana.
- Zamani Ashtiani S., **Sarabian, M.**, Laksari K., & Babaee H. (2022, November). *Blood flow predictions in data-poor regimes: A physics-informed Bayesian approach*. The 75th Annual Meeting of the American Physical Society's Division of Fluid Dynamics (APS-DFD), Indianapolis, Indiana.
- **Sarabian, M.**, Babaee, H., & Laksari, K. (2021, November). *Brain hemodynamic predictions from noninvasive Transcranial Doppler ultrasound and angiography data Using Physics-Informed Neural Networks*. The 74th Annual Meeting of the American Physical Society's Division of Fluid Dynamics (APS-DFD), Phoenix, Arizona.
- **Sarabian, M.**, Rosti, M.E., Brandt, L., & Hormozi, S. (2019, October). *Shear-induced sedimentation of a sphere in yield stress fluids: A computational study*. The 91th Annual Meeting of The Society of Rheology (SOR), Raleigh, North Carolina.
- **Sarabian, M.**, Metzger, B., & Hormozi, S. (2019, October). *Dynamics of shear-induced migration of non-Brownian suspension in a Large-gap Taylor Couette cell covered with bumpy rough surfaces*. The 91th Annual Meeting of The Society of Rheology (SOR), Raleigh, North Carolina.
- **Sarabian, M.**, Rosti, M.E., Brandt, L., & Hormozi, S. (2019, November). *Particle resolved simulations of a sphere settling in simple shear flows of yield-stress fluids*. The 72^{nd} Annual Meeting of the American Physical Society's Division of Fluid Dynamics (APS-DFD), Seattle, Washington.
- **Sarabian, M.**, Rosti, M.E., Brandt, L., & Hormozi, S. (2019, April). *From rheology to biology: Experiments & computations*. Ohio University Student Expo, Athens, Ohio.
- **Sarabian, M.**, Rosti, M.E., Brandt, L., & Hormozi, S. (2018, November). *Interface-resolved simulations of a sphere settling in simple shear flows of elastoviscoplastic fluids*. The 71^{nd} Annual Meeting of the American Physical Society's Division of Fluid Dynamics (APS-DFD), Atlanta, Georgia.
- Rashedi, A.R., **Sarabian, M.**, Ovarlez, G., & Hormozi, S. (2018, November). *An experimental study on fracturing flows of Newtonian fluids*. The 71^{nd} Annual Meeting of the American Physical Society's Division of Fluid Dynamics (APS-DFD), Atlanta, Georgia.
- **Sarabian, M.**, Rosti, M.E., Brandt, L., & Hormozi, S. (2018, April). *Computational modeling of confined elastoviscoplastic particulate flows under shear*. Ohio University Student Expo, Athens, Ohio.
- **Sarabian, M.**, Rosti, M.E., Brandt, L., & Hormozi, S. (2018, October). *Computational modeling of sheared yield-stress fluids*. Statewide Users Group (SUG) Conference, Ohio Supercomputer Center, Columbus, Ohio.
- **Sarabian, M.**, Metzger, B., & Hormozi, S. (2017, November). *Dispersion and layering of solid particles in cylindrical Couette flows*. The 70^{nd} Annual Meeting of the American Physical Society's Division of Fluid Dynamics (APS-DFD), Denver, Colorado.

Honors & Awards

2019	**Ohio University Student Research Expo Award**, Award offered from Ohio university in honor of being ranked 1^{st} among all graduate students in Mechanical Engineering department who participate in a competition judging session. This award is for student research and creativity activity at Ohio university.	*Ohio, U.S.A*
2018	**Research Article Featured with a Cover Image in *International Journal of Numerical Methods in Fluids (IJNMF)***, Our publication titled "Computational modeling of multiphase viscoelastic and elastoviscoplastic flows" is averaged 30% higher altmetric attention scores and 35% higher full text views on Wiley online library. It is featured and used on the cover of the journal.	*Ohio, U.S.A*
2017,2019	**Student Travel Award (TA)**, Award offered from graduate college at Ohio university to support participation in seminars, workshops and conferences.	*Ohio, U.S.A*
2015	**Scholarship**, Award offered by the department of Mechanical Engineering at Ohio university to support my education as a PhD student.	*Ohio, U.S.A*
2014	**Top Student Honor**, Ranked 1^{st} in the Mechanical Engineering; MS program at Shiraz University.	*Shiraz, Iran*
2014	**Outstanding Student Award**, Merit-based admission to PhD program as the top graduate student in Mechanical Engineering from Shiraz University.	*Shiraz, Iran*
2007	**Iranian National University Entrance Exam ("Konkur"), Mathematics & Physics track**, **Rank 52 of $\approx$500,000 candidates (Top 0.02%)**. (*Highly competitive, standardized exam for university admission; Math & Physics is the engineering pathway.*).	*Shiraz, Iran*

References

Dr. Kaveh Laksari **Postdoc Advisor**

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Dr. Bloen Metzger **Ph.D. Co-Advisor**

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Dr. Luca Brandt **Ph.D. Co-Advisor**

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